Fiscal Year:	FY 2018	**Task Last Updated:**	FY 05/17/2021
PI Name:	Furst, Eric Ph.D.		
Project Title:	Investigating the Structure of Paramagnetic Aggregates from Colloidal Ellipsoids-3 (InSPACE 3)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	furst@udel.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	
Organization Name:	University of Delaware		
PI Address 1:	Chemical & Biomolecular Engineering		
PI Address 2:	150 Academy Street		
PI Web Page:			
City:	Newark	**State:**	DE
Zip Code:	19716	**Congressional District:**	1
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	03/01/2010	**End Date:**	04/19/2018
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	McQuillen, John	**Contact Phone:**	216-433-2876
Contact Email:	john.b.mcquillen@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	InSPACE ISS		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX16AD21G ; NNX10AE44G ; NNX07AD02G (InSPACE2)		
Performance Goal No.:			
Performance Goal Text:			

InSPACE is a microgravity fluid physics experiment that was performed on the International Space Station (ISS). InSPACE began in December 1997 and concluded with the third series of follow-on InSPACE experiments aboard the ISS in November 2013.
[Ed. note April 2021: added to Task Book when received information. Note that this investigator's (Dr. Eric Furst) original grant was InSPACE-2: Structural and Rheological Transitions of Field-Responsive Fluids in Microgravity, grant NNX07AD02G; period of performance was 1/8/2007-1/07/2010.

Project continued with Dr. Furst as PI 3/1/2010-1/29/2016 as Investigating the Structure of Paramagnetic Aggregates from Colloidal Ellipsoids-3 (InSPACE 3), grant NNX10AE44G; 10/20/2015-4/19/2018, grant NNX16AD21G with

Task Description:	same title, Investigating the Structure of Paramagnetic Aggregates from Colloidal Ellipsoids-3 (InSPACE 3)] InSPACE-2 was a continuation of the InSPACE investigation begun on ISS Expedition 6, providing new and improved samples for operation in the Microgravity Science Glovebox (MSG). The use of external fields to control the microstructure of colloidal suspensions has long been recognized as a powerful means for tailoring the mechanical, optical, and electronic properties of materials. Magnetorheological (MR) suspensions, in particular, provide a striking example. These normally stable fluids undergo a dynamic transition to a solid within milliseconds after the application of an external magnetic field. They are also important models for developing methods of bottom-up fabrication of micro- and nano-structured materials and devices using field-directed self-assembly. Investigating the Structure of Paramagnetic Aggregates from Colloidal Emulsions-2 (InSPACE-2) experiments focused on the structure of magnetically-polarizable particle suspensions over long times in steady (DC) and pulsed magnetic fields. In pulsed fields especially, the long-time kinetics of the suspension micro-structural coarsening provided new and important results. The InSPACE-2 experiments were the first to yield information on the full, three-dimensional aggregation process over timescales much longer than those that are accessible on the ground, which are limited by catastrophic sedimentation. Furthermore, the experiments identified a novel and intriguing dynamic instability in which the suspension microstructures were observed to buckle at specific field frequencies and field strengths. For InSPACE-3, we will investigate the effect of particle shape on the micro-structural evolution of MR suspensions. Recent ground-based experiments demonstrate a startling effect that particle shape has on the interactions of dipolar chains and the resulting suspension microstructure. Specifically, we will investigate suspensions of paramagnetic ellipsoid-shaped particles. In combination with the results of InSPACE-2, we hypothesize that particle shape will dramatically alter the aggregation kinetics, microstructures, and microscopic mechanics. This will potentially lead to the ability to engineer enhanced properties of these suspensions, including suppression of the lateral aggregation in magnetorheological fluid-based electromechanical devices or the ability to create new colloidal materials through field-directed self-assembly.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Space Applications: The quick phase-shifting property of magnetic colloidal fluids makes them potentially useful for devices such as optical interfaces, active noise, and vibration dampers inside spacecrafts. Future uses could include robotics, energy-transfer devices such as clutches, and other control systems. Earth Applications: The work has application to directed self-assembly of crystalline structures from particles that may eventually allow creation of new nano-materials fabricated from nanoparticle building blocks, with potential applications in medicine, energy storage, chemical separations, and catalysis. Magnetic colloidal fluid technology is currently being used for shock absorbers in racecars and sport cars. This technology is expanding to include making large-scale building foundation stabilizers for areas prone to earthquakes.
Task Progress:	See Bibliography section below for publications resulting from this NASA research. [Ed., May 2021]
Bibliography Type:	Description: (Last Updated: 01/27/2026)
Articles in Peer-reviewed Journals	Kim H, Bauer JL, Vasquez PA, Furst EM. "Structural coarsening of magnetic ellipsoid particle suspensions driven in toggled fields." Journal of Physics D: Applied Physics. 2018 May;52(18):184002. https://doi.org/10.1088/1361-6463/ab062f , May-2018
Articles in Peer-reviewed Journals	Kang F, Kim H, Gray LAG, Christie D, Jonas U, Graczykowski B, Furst EM, Priestley RD, Fytas G. "Ultrathin shell layers dramatically influence polymer nanoparticle surface mobility." Macromolecules 2018 Nov 13;51(21):8522-9. https://doi.org/10.1021/acs.macromol.8b01804 ; PMID: 30906073; PMCID: PMC6428372 , Nov-2018
Articles in Peer-reviewed Journals	Kim H, Cang Y, Kang E, Graczykowski B, Secchi M, Montagna M, Priestley RD, Furst EM, Fytas G. "Direct observation of polymer surface mobility via nanoparticle vibrations." Nat Commun. 2018 Jul 25;9(1):2918. https://doi.org/10.1038/s41467-018-04854-w ; PMID: 30046038; PMCID: PMC6060150 , Jul-2018
Articles in Peer-reviewed Journals	Bauer JL, Kurian MJ, Stauffer J, Furst EM. "Suppressing the Rayleigh-Plateau instability in field-directed colloidal assembly." Langmuir. 2016 Jul 5;32(26):6618-23. https://doi.org/10.1021/acs.langmuir.6b00771 , Jul-2016
Journal/Magazine covers	Bauer JL, Kurian MJ, Stauffer J, Furst EM. "Cover in Langmuir for article, 'Suppressing the Rayleigh-Plateau instability in field-directed colloidal assembly. Langmuir. 2016 Jul 5;32(26):6618–23.' Cover published in issue 27." Langmuir. 2016 Jul 12;32(27). Cover image: https://pubs.acs.org/toc/langd5/32/27 , Jul-2016